

Mammalia Raccoon Proximity Network Analysis – Project Explanation

Project Overview

This project analyzes **social interaction patterns among urban raccoons** using **network analysis** to understand how **rabies and other pathogens can spread** within raccoon populations.

Since raccoons are a major **vector of rabies**, studying how often and how closely they interact helps estimate **disease transmission risk** and supports better **public health and wildlife management strategies**.

Objective

- Model raccoon interactions as a **graph/network**
- Identify **high-risk raccoons** that can spread pathogens quickly
- Use **centrality measures** to understand social structure and disease flow

Dataset & Network Construction

- **24 raccoons** → represented as **nodes**
- **~2000 interactions** → represented as **weighted edges**
- Data collected using **proximity-detecting collars**
- Edge **weight = time spent in close proximity**
- Monthly interaction data used to build the network

👉 If two raccoons are connected, it means they had **close physical interaction**, which is critical for rabies transmission.

Why Network Analysis?

Disease spread is not random.

It depends on:

- Who interacts with whom
- How frequently
- Who acts as a bridge between groups

Network analysis helps identify **super-spreaders** and **critical connectors**.

Centrality Measures Used & Interpretation

1. Degree Centrality

- Measures **number of direct connections**
- High degree = more interactions

Inference:

- **Raccoon #4** has the highest degree
- Acts as a **high-contact individual**
- High risk of **initiating widespread infection**

2. Closeness Centrality

- Measures how **close a node is to all others**
- Indicates how fast something can spread

Inference:

- **Raccoon #2** has highest closeness centrality
- Can **spread pathogens rapidly** across the entire network

3. Betweenness Centrality

- Measures how often a node lies on **shortest paths**
- Identifies **bridge nodes**

Inference:

- **Raccoon #16** connects different subgroups
- Acts as a **key transmission bridge**
- Removing or vaccinating this raccoon can significantly **slow disease spread**

4. Clustering Coefficient

- Measures how tightly a raccoon is embedded in a **local group**
- Indicates **family or group behavior**

Inference:

- **Raccoon #2** has highest clustering
- Infection here may cause **localized outbreaks within families**

5. Eigenvector Centrality

- Measures influence based on **connections to influential nodes**

Inference:

- **Raccoon #4** again ranks highest
- Connected to other highly interactive raccoons
- Considered a **super-spreader candidate**

6. PageRank Algorithm

- Evaluates overall importance considering **direction and weight of interactions**
- Helps rank raccoons by **global influence**

Key Findings

- Raccoon populations are **highly interconnected**
- Disease can spread **much faster than expected**
- A few raccoons play **disproportionately large roles** in pathogen transmission
- Targeted intervention (vaccination/tracking of key raccoons) is more effective than random control

Technologies & Concepts Used

- Graph Theory
- NetworkX (Python)
- Weighted graphs

- Centrality algorithms
- Epidemiological modeling concepts